Band-1 Computational Experiment: Sierpiński Problem — Cover Verification for k=78557

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note verifies the **classical Selfridge covering** that proves $k = 78557$ is a Sierpiński number: for every $n \geq 1$, $k \cdot 2^n + 1$ has a small prime factor from a fixed finite set. It does **not** claim that all Sierpiński numbers are classified for all k; it does **not** claim Clay / ToE; remaining Seventeen-or-Bust candidates stay **OPEN**. Finite cover verification $\neq$ full classification.

Abstract

A **Sierpiński number** is an odd positive integer k such that $k \cdot 2^n + 1$ is composite for every integer $n \geq 1$. Sierpiński (1960) proved infinitely many such k exist. Selfridge (1962) showed that

$$k = 78557 = 17 \cdot 4621$$

is Sierpiński via the covering set

$$C = \{3, 5, 7, 13, 19, 37, 73\}$$

with covering modulus $M = 36$: for each residue class of n modulo 36, a fixed $p \in C$ divides $k \cdot 2^n + 1$.

Box result: the classical residue→factor table is re-checked modularly; orders of 2 modulo each $p \in C$ all divide 36; lcm = 36; every residue r mod 36 is covered; 7199 systematic spot-checks and 500 random $n \leq 10^6$ all pass. **Cover verified: YES.** Wall ≈ 0.021 s.

Status of the smallest-k conjecture: OPEN — five candidates $k \in \{21181, 22699, 24737, 55459, 67607\}$ remain (Keller / PrimeGrid, cited 2026).

Problem statement

Sierpiński problem. Determine the smallest Sierpiński number.

This experiment. Independently verify Selfridge's cover for the known smallest *candidate* $k = 78557$ (already proved Sierpiński; conjectured minimal). Optionally ledger other known small Sierpiński k and remaining SoB candidates from the literature.

Method (box)

1. **Covering set / modulus.** $C = \{3, 5, 7, 13, 19, 37, 73\}$, $M = 36$.
2. **Orders.** Compute $\mathrm{ord}_p(2)$ for each $p \in C$; assert each divides M and lcm $= M$. Then $2^n \equiv 2^r \pmod{p}$ whenever $n \equiv r \pmod{M}$.
3. **Residue table.** For each $r = 0, \ldots, 35$, check $k \cdot 2^r + 1 \equiv 0 \pmod{p}$ for the classical factor $p = p(r)$, and record all hits in C.
4. **Propagation spot-checks.** For $n = r + 36t$ ($t = 0..199$, $n \geq 1$) and 500 random $n \leq 10^6$, assert the classical $p(r)$ divides $k \cdot 2^n + 1$ via modular exponentiation.
5. **No claim** about k outside the verified cover, nor about remaining SoB k.

Classical residue → factor table (Selfridge)

n mod 36	0	1	2	3	4	5	6	7	8	9	10	11
factor	3	5	3	73	3	5	3	7	3	5	3	13

n mod 36	12	13	14	15	16	17	18	19	20	21	22	23
factor	3	5	3	19	3	5	3	7	3	5	3	13

n mod 36	24	25	26	27	28	29	30	31	32	33	34	35
factor	3	5	3	37	3	5	3	7	3	5	3	13

Results (numbers!)

Unit / structural checks

Check	Result
$\mathrm{ord}_3(2)$	2
$\mathrm{ord}_5(2)$	4
$\mathrm{ord}_7(2)$	3
$\mathrm{ord}_{13}(2)$	12
$\mathrm{ord}_{19}(2)$	18
$\mathrm{ord}_{37}(2)$	36
$\mathrm{ord}_{73}(2)$	9
lcm of orders	$36 = M$
Residues 0..35 with ≥ 1 cover factor	36/36
Classical table matches modular arithmetic	**PASS**
Spot-checks $n = r + 36t$ (7199 values)	**PASS**
Random $n \leq 10^6$ (500)	**PASS**
Cover verified	**YES**
Wall	≈ 0.021 **s**

Small-n samples (exact $N = k \cdot 2^n + 1$)

For $n = 1..24$, the classical factor divides the integer N exactly (see `results.json` `small_N_samples`). Example: $n = 1$, $N = 157115 = 5 \cdot 7 \cdot 4489$; $n = 3$, $N = 628457 = 73 \cdot 8609$.

Literature ledger (not re-proved on box)

Other known small Sierpiński k (covers cited; not re-verified tonight)

k	Covering set (literature)	Note
78557	$\{3, 5, 7, 13, 19, 37, 73\}$	Selfridge 1962; **box-verified**
271129	$\{3, 5, 7, 13, 17, 241\}$	Smallest known **prime** Sierpiński
271577	$\{3, 5, 7, 13, 17, 241\}$	
322523	$\{3, 5, 7, 13, 37, 73, 109\}$	
327739	$\{3, 5, 7, 13, 17, 97, 257\}$	
482719	$\{3, 5, 7, 13, 17, 241\}$	
575041, 603713, 903983, 965431, ...	various	see `literature_ledger.json`

Sources: Prime Glossary (t5k.org); Keller, *The Sierpiński Problem* (http://www.prothsearch.com/sierp.html).

Remaining candidates (original Sierpiński / Seventeen or Bust)

To prove 78557 is the **smallest** Sierpiński number, one must exhibit a prime $k \cdot 2^n + 1$ for every odd $k < 78557$. As of Keller's compilation (updated **2026-07-29**) and PrimeGrid SoB status, **five** k remain:

$$k \in \{21181,\ 22699,\ 24737,\ 55459,\ 67607\}.$$

Status: OPEN. No covering set is known for these; distributed search continues (PrimeGrid leading edge cited $\sim 4.7 \times 10^7$). Conjecture: each eventually yields a prime, so none is Sierpiński — but that is **not proved**.

Honesty / non-claims

- **Does:** verify Selfridge's finite cover for $k = 78557$ on-box.
- **Does not:** classify all Sierpiński k; settle the five remaining SoB candidates; claim Clay Millennium or ToE relevance; claim new Sierpiński numbers.

Band-4 / OPEN: "Is 78557 the smallest Sierpiński number?"

Reproducibility

`python3 /workspace/experiments/band1/sierpinski/verify_cover_78557.py`

Artifacts: `results.json`, `SUMMARY.txt`, `literature_ledger.json`, this note.

References

1. W. Sierpiński, *Sur un problème concernant les nombres $k \cdot 2^n + 1$*, Elem. Math. **15** (1960), 73–74.
2. J. L. Selfridge (1962), covering proof for $k = 78557$ (via Erdős correspondence / standard citation).
3. W. Keller, *The Sierpiński Problem*, http://www.prothsearch.com/sierp.html (updated 2026-07-29).
4. Prime Glossary: Sierpinski number — https://t5k.org/glossary/page.php?sort=SierpinskiNumber
5. Seventeen or Bust / PrimeGrid — remaining k search.